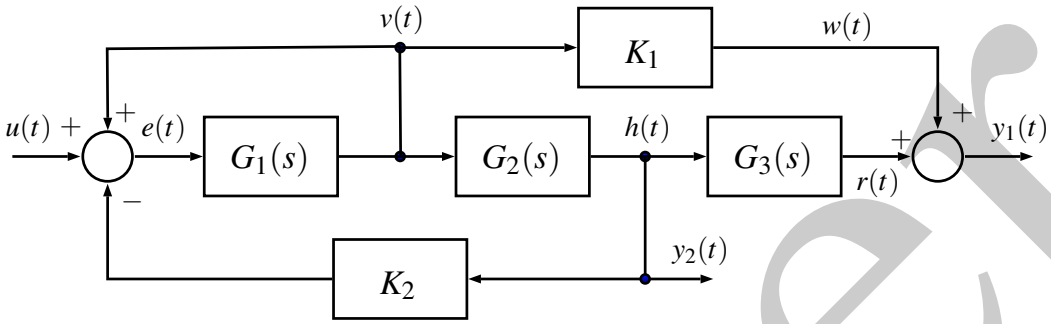
SOLUTIONS: CONTROL SYSTEMS – 2025/2026

1. Solution

a) In figure below, we report the block scheme shown in Figure 1.1 in the text of the exam paper where, with some abuse of notation, we represent the inputs and outputs of each block in the time-domain instead of in the Laplace domain:



By inspecting the above block scheme, and referring to the input $u(t)$, to the outputs $y_1(t)$ and $y_2(t)$, and to the "intermediate" variables $e(t)$, $v(t)$, $h(t)$, $w(t)$, and $r(t)$, the following equations are easily obtained (we drop the dependence on t for simplicity):

$$\begin{cases} e = v + u - K_2 \cdot h \\ h = G_2 \cdot v \\ v = G_1 \cdot e \\ w = K_1 \cdot v \\ r = G_3 \cdot h \\ y_1 = r + w \\ y_2 = h \end{cases}$$

Hence:

$$e = G_1 \cdot e + u - K_2 \cdot G_1 \cdot G_2 \cdot e \Longrightarrow (1 - G_1 + K_2 \cdot G_1 \cdot G_2) \cdot e = u$$

which gives

$$e = \frac{1}{1 - G_1 + K_2 \cdot G_1 \cdot G_2} \cdot u$$

Now:

$$y_1 = G_1 \cdot G_2 \cdot G_3 \cdot e + K_1 \cdot G_1 \cdot e = (G_1 \cdot G_2 \cdot G_3 + K_1 \cdot G_1) \cdot e$$

thus getting

$$y_1 = \frac{G_1 \cdot G_2 \cdot G_3 + K_1 \cdot G_1}{1 - G_1 + K_2 \cdot G_1 \cdot G_2} \cdot u = \frac{G_1 \cdot (G_2 \cdot G_3 + K_1)}{1 - G_1 \cdot (1 - K_2 \cdot G_2)} \cdot u$$

Thus, for $K_1 \neq 0$ and $K_2 \neq 0$, it follows that

$$G_{u,y_1}(s) = \frac{G_1(s) \cdot [G_2(s) \cdot G_3(s) + K_1]}{1 - G_1(s) \cdot [1 - K_2 \cdot G_2(s)]}$$

[10 Marks]

Answers

b) From the above answer to Question 1-a), we obtain immediately

$$y_2 = h = G_1 \cdot G_2 \cdot e = \frac{G_1 \cdot G_2}{1 - G_1 \cdot (1 - K_2 \cdot G_2)} \cdot u$$

Thus, for $K_1 \neq 0$ and $K_2 \neq 0$, it follows that

$$G_{u,y_2}(s) = \frac{G_1(s) \cdot G_2(s)}{1 - G_1(s) \cdot [1 - K_2 \cdot G_2(s)]}$$

[5 Marks]

c) Plugging $K_1 = 0$ and $K_2 = 0$ into the expression of $G_{u,y_1}(s)$ obtained in the answer to Question 1-a), it follows immediately that

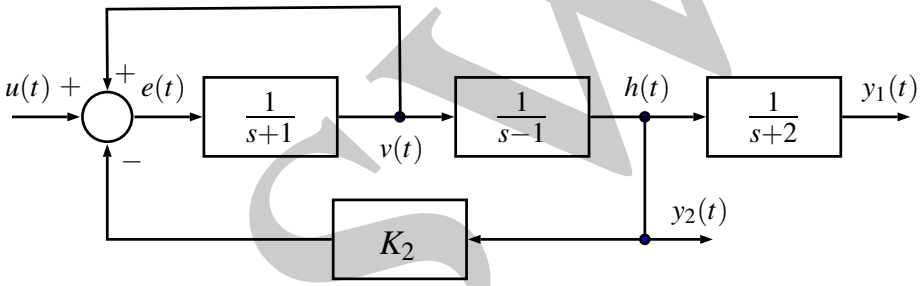
$$G_{u,y_1}(s) = \frac{G_1(s) \cdot G_2(s) \cdot G_3(s)}{1 - G_1(s)}$$

[3 Marks]

d) From the text of Question 1-d), we have $K_1 = 0$ and

$$G_1(s) = \frac{1}{s+1}; \quad G_2(s) = \frac{1}{s-1}; \quad G_3(s) = \frac{1}{s+2}$$

Therefore, the block scheme shown in Figure 1.1 in the text of the exam paper reduces to the block scheme depicted in figure below:



i) Consider the three blocks shown in the block scheme in figure above with outputs $v(t)$, $h(t)$, and $y_1(t)$, and transfer functions $\frac{1}{s+1}$, $\frac{1}{s-1}$ and $\frac{1}{s+2}$, respectively. Clearly only these three blocks represent dynamic systems and hence being described via the introduction of state variables.

We proceed by defining the state-space representations associated to each of these three blocks using one of the possible realisation procedures. In general, given a strictly proper transfer function of the form

$$G(s) = \frac{\beta_m s^m + \beta_{m-1} s^{m-1} + \cdots + \beta_1 s + \beta_0}{s^n + \alpha_{n-1} s^{n-1} + \cdots + \alpha_1 s + \alpha_0} \quad \text{with} \quad m < n$$

Answers

a possible state-space realisation is given by

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -\alpha_0 \\ 1 & 0 & \cdots & 0 & -\alpha_1 \\ \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & -\alpha_{n-2} \\ 0 & \cdots & 0 & 1 & -\alpha_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \\ 0 \\ \vdots \\ 0 \end{bmatrix} u \\ \\ y = \begin{bmatrix} 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \end{cases}$$

Therefore, a *possible choice* for the state variables is:

$$x_1(t) := v(t); \quad x_2(t) := h(t); \quad x_3(t) := y_1(t)$$

and from the above realisation form, we obtain the following three state-space representations:

$$G_1(s) = \frac{1}{s+1} \Longrightarrow \begin{cases} \dot{x}_1 = -x_1 + e \\ v = x_1 \end{cases}$$

$$G_2(s) = \frac{1}{s-1} \Longrightarrow \begin{cases} \dot{x}_2 = x_2 + v \\ h = x_2 \end{cases}$$

$$G_3(s) = \frac{1}{s+2} \Longrightarrow \begin{cases} \dot{x}_3 = -2x_3 + h \\ y_1 = x_3 \end{cases}$$

Hence, we define the state vector:

$$x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

Thus, by inspecting the block scheme in the above figure we obtain:

$$\begin{cases} \dot{x}_1(t) = -K_2 x_2(t) + u(t) \\ \dot{x}_2(t) = x_1(t) + x_2(t) \\ \dot{x}_3(t) = x_2(t) - 2x_3(t) \\ y_1(t) = x_3(t) \\ y_2(t) = x_2(t) \end{cases}$$

Accordingly, matrices A, B, C, D of the state space description are given by:

$$A = \begin{bmatrix} 0 & -K_2 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -2 \end{bmatrix}, \; B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \; C = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \; D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

[12 Marks]

Answers

ii) The stability properties of the dynamic system represented by the state equations obtained in the answer to Question 1-d)-i) as a function of the scalar $K_2 \in \mathbb{R}$ only depend on the state matrix A of any state-space description of the system. From the answer to Question 1-d)-i), we have:

$$A = \begin{bmatrix} 0 & -K_2 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

which **does depend on** K_2. Therefore, the stability properties also depend on K_2. The characteristic polynomial is:

$$\phi_A(s) = \det(sI - A) = \det \begin{bmatrix} s & K_2 & 0 \\ -1 & s-1 & 0 \\ 0 & -1 & s+2 \end{bmatrix}$$

and hence

$$\phi_A(s) = (s+2)\,[s(s-1)+K_2] = (s+2)(s^2 - s + K_2)$$

Thus, it is immediate to conclude that for every value of $K_2 \in \mathbb{R}$ the system is unstable because matrix A has at least one eigenvalue with positive real part.

[5 Marks]

iii) Consider the expression of $G_{u,y_1}(s)$ obtained in the answer to Question 1-a), that is:

$$G_{u,y_1}(s) = \frac{G_1(s) \cdot [G_2(s) \cdot G_3(s) + K_1]}{1 - G_1(s) \cdot [1 - K_2 \cdot G_2(s)]}$$

and substitute $K_1 = 0$ and

$$G_1(s) = \frac{1}{s+1}; \quad G_2(s) = \frac{1}{s-1}; \quad G_3(s) = \frac{1}{s+2}$$

into $G_{u,y_1}(s)$ thus obtaining

$$G_{u,y_1}(s) = \frac{G_1(s) \cdot [G_2(s) \cdot G_3(s) + K_1]}{1 - G_1(s) \cdot [1 - K_2 \cdot G_2(s)]} = \frac{\dfrac{1}{(s+1)(s-1)(s+2)}}{1 - \dfrac{1}{s+1}\left(1 - \dfrac{K_2}{s-1}\right)}$$

Simple algebra finally yields

$$G_{u,y_1}(s) = \frac{1}{(s+2)(s^2 - s + K_2)}$$

Hence, the denominator of $G_{u,y_1}(s)$ coincides with the characteristic polynomial $\phi_A(s)$ obtained in the answer to Question 1-d)-ii) which implies that the poles of $G_{u,y_1}(s)$ coincide with the eigenvalues of matrix A. Therefore, the dynamic system represented by $G_{u,y_1}(s)$ is unstable for every value of $K_2 \in \mathbb{R}$ because $G_{u,y_1}(s)$ has at least one pole with positive real part.

[5 Marks]

Answers

2. Solution

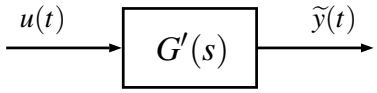
a) i) Consider the transfer function

$$G(s) = 4 \cdot \frac{3e^{-s}}{9+6s+s^2}$$

and let us rewrite it as

$$G(s) = G'(s) \cdot e^{-s} \quad \text{where} \quad G'(s) = \frac{12}{9+6s+s^2}$$

The multiplicative term e^{-s} represents a delay of 1 s in the time-domain and, as such, it does not influence the existence of a sinusoidal regime (e.g., steady-state) output function $y_{\text{reg}}(t), t \in \mathbb{R}$. Therefore, to establish the existence of a sinusoidal regime we only focus on the system shown in figure below in which $\widetilde{y}(t)$ represents the output neglecting the delay term.



Since

$$G'(s) = \frac{12}{9+6s+s^2} = \frac{12}{(s+3)^2}$$

it follows that the poles of $G'(s)$ have strictly negative real part and therefore the dynamic system is asymptotically stable. As a consequence, the frequency response theorem can be applied which implies the existence of a sinusoidal regime $y_{\text{reg}}(t), t \in \mathbb{R}$ when the system is subjected to the sinusoidal input $u(t) = [\sin(3t)] \cdot \mathbf{1}(t)$ considered in Question 2-a)-i).

[4 Marks]

ii) Owing to the frequency response theorem (that can be used as per the answer to Question 2-a)-i)), the analytical expression of the output function regime (e.g., steady-state) output function $y_{\text{reg}}(t), t \in \mathbb{R}$ is given by

$$y_{\text{reg}}(t) = |G(j3)| \sin[3t + \arg G(j3)]$$

where

$$|G(j3)| = 12 \cdot \frac{\left|e^{-j3}\right|}{\left|(j3+3)^2\right|} = \frac{2}{3}$$

$$\arg G(j3) = -2\arg(j3+3) - 3\frac{180^\circ}{\pi} \simeq -262^\circ$$

[5 Marks]

b) i) Consider the input function:

$$u(t) = \left(1-e^{-t}\right)\mathbf{1}(t) - \left[1-e^{-(t-7)}\right]\mathbf{1}(t-7)$$

where $\mathbf{1}(t)$ denotes the unit step function. Let us first rewrite the input as follows:

$$u(t) = u_1(t) - u_2(t)$$

where

$$u_1(t) = \left(1-e^{-t}\right)\mathbf{1}(t)$$

Answers

and

$$u_2(t) = \left[1 - e^{-(t-7)}\right] \mathbf{1}(t-7)$$

By applying the Laplace transform properties, we obtain:

$$U(s) = U_1(s) - U_2(s)$$

where

$$U_1(s) = \mathscr{L}[u_1(t)] = \frac{1}{s} - \frac{1}{s+1}$$

and

$$U_2(s) = \mathscr{L}[u_2(t)] = \left(\frac{1}{s} - \frac{1}{s+1}\right) \cdot e^{-7s}$$

Hence:

$$Y(s) = \mathscr{L}[y(t)] = \frac{12}{(s+3)^2} \cdot e^{-s} \left[U_1(s) - U_2(s)\right]$$

yielding

$$Y(s) = \frac{12}{(s+3)^2} \cdot \left(\frac{1}{s} - \frac{1}{s+1}\right) \cdot (e^{-s} - e^{-8s})$$

To simplify the subsequent calculations, we are going to exploit the linearity, time-invariance, and time-translation properties. In this respect, we consider a new auxiliary input

$$\widetilde{u}(t) = \left(1 - e^{-t}\right) \mathbf{1}(t)$$

and its Laplace transform

$$\widetilde{U}(s) = \mathscr{L}[\mathbf{1}(t)] = \frac{1}{s} - \frac{1}{s+1}$$

The Laplace transform of the output generated by this auxiliary input is given by:

$$\widetilde{Y}(s) = \frac{12}{(s+3)^2} \cdot \left(\frac{1}{s} - \frac{1}{s+1}\right) = \frac{12}{s(s+1)(s+3)^2}$$

Proceeding by partial fractions expansion we get:

$$\widetilde{Y}(s) = \frac{\alpha}{s} + \frac{\beta}{s+1} + \frac{\gamma}{s+3} + \frac{\delta}{(s+3)^2}$$

After some algebraic calculations, one gets:

$$\frac{12}{s(s+1)(s+3)^2} = \\ = \frac{(\alpha+\beta+\gamma)s^3 + (7\alpha+6\beta+4\gamma+\delta)s^2 + (15\alpha+9\beta+3\gamma+\delta)s + 9\alpha}{s(s+1)(s+3)^2}$$

and hence after further calculations, eventually we get

$$\alpha = \frac{4}{3};\ \beta = -3;\ \gamma = \frac{5}{3};\ \delta = 2$$

Thus:

$$\widetilde{Y}(s) = \frac{4}{3} \cdot \frac{1}{s} - 3 \cdot \frac{1}{s+1} + \frac{5}{3} \cdot \frac{1}{s+3} + 2 \cdot \frac{1}{(s+3)^2}$$

Answers

which gives

$$\widetilde{y}(t) = \mathscr{L}^{-1}[\widetilde{Y}(s)] = \left[\frac{4}{3} - 3e^{-t} + \frac{5}{3}e^{-3t} + 2te^{-3t}\right]\mathbf{1}(t)$$

Finally, owing to the above-mentioned linearity, time-invariance, and time-translation properties, we finally obtain:

$$\begin{aligned} y(t) &= \mathscr{L}^{-1}[Y(s)] = \\ &= \left[\frac{4}{3} - 3e^{-(t-1)} + \frac{5}{3}e^{-3(t-1)} + 2(t-1)e^{-3(t-1)}\right]\mathbf{1}(t-1) \\ &\quad - \left[\frac{4}{3} - 3e^{-(t-8)} + \frac{5}{3}e^{-3(t-8)} + 2(t-8)e^{-3(t-8)}\right]\mathbf{1}(t-8) \end{aligned}$$

[8 Marks]

ii) Figure 2.1 below shows the time behaviour of the input $u(t)$ and the output $y(t)$.

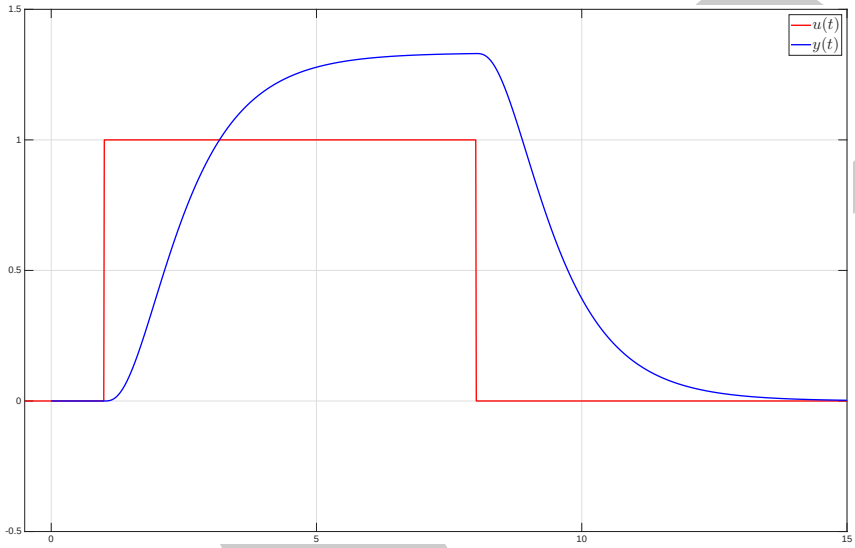


Figure 2.1 Time behaviour of the input $u(t)$ and the output $y(t)$.

[3 Marks]

Answers

3. Solution

a) As requested in Question 3 of the exam paper, the controller $C(s)$ to be designed is constrained to take on the form

$$C(s) = \mu_C \frac{1+s\tau}{s}$$

where $\mu_C > 0$ and $\tau > 0$ are the two parameters to be determined so that Requirements i), ii), and iii) are simultaneously satisfied.

Let us first deal with Requirement i) that refers to the closed-loop steady-state error $\bar{e}_\infty$ generated by a "ramp-like" input of the form

$$w(t) = A \cdot t \cdot \mathbf{1}(t)\,,\ \forall A \text{ such that } |A| \le 1 \quad \Longrightarrow \quad |\bar{e}_\infty| \le 0.5$$

where $\mathbf{1}(t)$ denotes the unit step function.

As initial attempt, set $\tau = 0$ thus using a controller of the form

$$C(s) = \mu_C \frac{1}{s}$$

Hence, the open-loop transfer function $L_1(s)$, also taking into account the time-delay block shown in Fig. 3.1 in the text of the exam paper is given by

$$L_1(s) = 12\,\mu_C \cdot \frac{1+0.1\,s}{s(1+24\,s)} e^{-0.1s}$$

Since Requirement i) refers to the closed-loop steady-state error $\bar{e}_\infty$, the effect of the time-delay can be neglected provided that the closed-loop system is asymptotically stable. In this respect, within the scope of Requirement i), we consider the open-loop transfer function

$$\widehat{L}_1(s) = 12\,\mu_C \cdot \frac{1+0.1\,s}{s(1+24\,s)}$$

The transfer function from the input $w(t)$ to the error $e(t)$ (neglecting the time-delay) is the sensitivity transfer function that reads

$$S(s) = \frac{1}{1+\widehat{L}_1(s)} = \frac{1}{1+12\,\mu_C \cdot \dfrac{1+0.1\,s}{s(1+24\,s)}} = \frac{s(1+24\,s)}{s(1+24\,s)+12\,\mu_C(1+0.1\,s)}$$

Thus, using the Final Value Theorem, it follows that

$$\bar{e}_\infty = \lim_{s\to 0} s \frac{s(1+24\,s)}{s(1+24\,s)+12\,\mu_C(1+0.1\,s)} \cdot \frac{A}{s^2} = \frac{A}{12\mu_C}$$

Therefore, Requirement i) is satisfied if the controller gain μ_C is selected so as

$$\frac{1}{12\mu_C} \le 0.5 \Longrightarrow \mu_C \ge \frac{1}{6} \simeq 0.17$$

We choose $\mu_C = 0.5$. Then

$$\widehat{L}_1(s) = 6 \cdot \frac{1+0.1\,s}{s(1+24\,s)}$$

To get insight into the constraints that Requirements ii) and iii) impose in the frequency domain, let us sketch the straight-line approximation of the Bode

Answers

diagram and a few points of the true Bode diagram of the open-loop transfer function $\widehat{L}_1(s)$.

The parameters of $\widehat{L}_1(s)$ are:

$$\mu = 6, \quad g = 1$$

and the relevant frequencies are:

$$\omega_z = 10, \quad \omega_{p_1} = 0, \quad \omega_{p_2} = 1/24\,.$$

- When $0 < \omega < 1/24$, the slope is -20 dB/dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$|\widehat{L}_1|_{\text{dB}} = -20\log(\omega) + 20\log(6)\,.$$

For $\omega = 1/24$, the magnitude is

$$|\widehat{L}_1|_{\text{dB}} = -20\log(1/24) + 20\log(6) \simeq 43.2\text{dB}\,.$$

- When $1/24 \leq \omega \leq 10$, the slope is -40 dB/dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$|\widehat{L}_1|_{\text{dB}} = -40\log(\omega) + 40\log(1/24) + 43.2\text{dB}\,.$$

For $\omega = 10$, the magnitude is

$$|\widehat{L}_1|_{\text{dB}} = -40\log(10) + 40\log(1/24) + 43.2\text{dB} \simeq -52\text{dB}\,.$$

- When $\omega > 10$, the slope is -20dB/dec. Thus the equation of the straight-line approximation of the magnitude is given by:

$$\begin{aligned} |\widehat{L}_1|_{\text{dB}} &= -20\log\omega + 20\log 10 - 52\text{dB} \\ &\simeq -20\log\omega - 32\text{dB}\,. \end{aligned}$$

For $\omega = 20$, the magnitude is

$$\begin{aligned} |\widehat{L}_1|_{\text{dB}} &= -20\log 20 - 32\text{dB} \\ &\simeq -58\text{dB}\,. \end{aligned}$$

Concerning the phase, we write:

$$\arg\widehat{L}_1(j\omega) = \arg(1 + 0.1\,j\omega) - 90^\circ - \arg(1 + 24\,j\omega)$$

We calculate $\arg\widehat{L}_1(j\omega)$ for $\omega = 1/24, 10$ thus getting:

$$\arg\widehat{L}_1(j/24) \simeq -134.8^\circ; \arg\widehat{L}_1(j10) \simeq -134.8^\circ$$

The resulting true Bode diagrams with their straight-line approximation are plotted in Figure 3.1 below.

Let us now focus on Requirement iii). Specifically, suppose that the phase margin φ_m satisfies $\varphi_m > 75^\circ$. Thus, we can approximately assume that there is only one real dominant pole corresponding to the cross-over frequency ω_c and then we can use the following approximation for the settling time $t_{s,0.01}$ at 1%:

$$t_{s,0.01} \simeq \frac{4.6}{\omega_c}$$

Answers

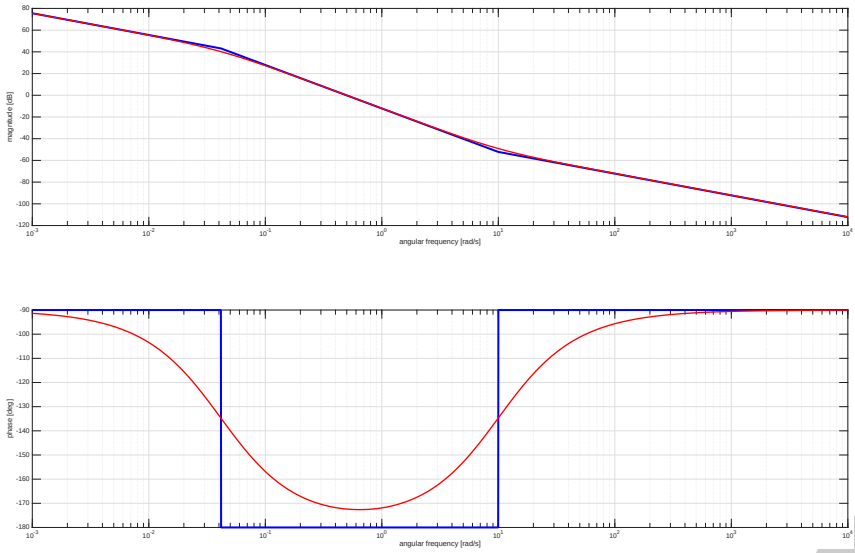


Figure 3.1 Bode diagrams of $\widehat{L}_1(j\omega)$ with $\mu_C = 0.5$, $\tau = 0$.

Therefore, Requirement iii) implies

$$\frac{4.6}{\omega_c} \leq 2 \Longrightarrow \omega_c \geq 4.6/2 = 2.3\text{rad/s}$$

We choose $\omega_c = 5\text{rad/s}$ to meet Requirement iii).

Now:

$$\arg\widehat{L}_1(j\omega_c) = \arg\widehat{L}_1(j5) \simeq -153^\circ$$

Hence, Requirement ii) on the phase margin **is not satisfied** because

$$\varphi_m \simeq 27^\circ$$

Indeed, taking into consideration the time-delay that has been neglected so far, we obtain:

$$\begin{aligned}\arg L_1(j\omega_c) &= \arg L_1(j5) \\ &= \arg(1+0.1j5) - 90^\circ - \arg(1+24j5) - \frac{180}{\pi}\cdot\frac{1}{10} \simeq -181^\circ < 0\end{aligned}$$

thus getting a **negative phase margin** and hence the system is **unstable** in closed-loop!

Therefore, we need to carry out a **second attempt** by designing the parameter $\tau > 0$ in the controller

$$\widehat{C}(s) = 0.5 \cdot \frac{1+s\tau}{s}$$

so as to provide a suitable increase of the phase in correspondence of the crossover frequency $\omega_c = 5\text{rad/s}$.

In this connection, as a second attempt, we select $\tau = 24$ so as to "cancel" the low-frequency open-loop pole thus getting

$$\widehat{C}(s) = 0.5 \cdot \frac{1+24s}{s}$$

The new open-loop transfer function $\widehat{L}_2(s)$ (still neglecting, for the time-being, the time-delay) reads

$$\widehat{L}_2(s) = 0.5 \cdot \frac{1+24s}{s} \cdot 12 \cdot \frac{1+0.1s}{1+24s} = 6 \cdot \frac{1+0.1s}{s}$$

Answers

The resulting true Bode diagrams with their straight-line approximation are plotted in Figure 3.2 below.

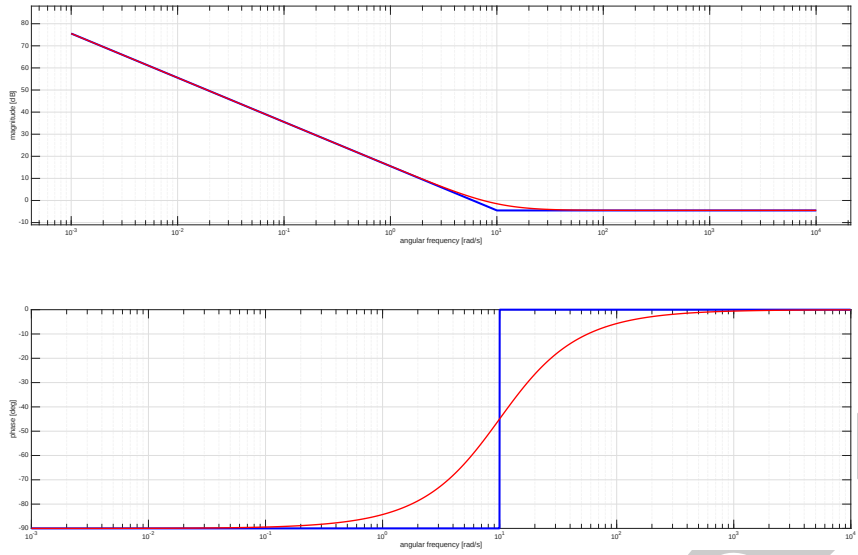


Figure 3.2 Bode diagrams of $\widehat{L}_2(j\omega)$ with $\mu_C = 0.5$, $\tau = 24$.

The inspection of the diagrams in Fig. 3.2 clearly show that the new design satisfies all requirements. Let us verify this claim.

First, let us compute the actual crossover frequency ω_c:

$$|\widehat{L}_2(j\omega_c)| = 1 \Longrightarrow 6 \cdot \frac{|1+0.1\,j\omega_c|}{|j\omega_c|} = 1$$

$$\Longrightarrow \sqrt{1+\frac{1}{100}\omega_c^2} = \frac{1}{6}\omega_c \Longrightarrow \omega_c = \frac{1}{\sqrt{\frac{1}{36}-\frac{1}{100}}} = \frac{15}{2} = 7.5\,\text{rad/s}$$

Hence, the settling time satisfies Requirement iii):

$$t_{s,0.01} \simeq \frac{4.6}{\omega_c} = \frac{4.6}{7.5} = 0.6\,\text{s} < 2\,\text{s}$$

Let us now check that also Requirement ii) is satisfied also taking into account the effect of the time-delay. Thus, we consider

$$L_2(s) = \widehat{L}_2(s) \cdot e^{-0.1s} = 6 \cdot \frac{1+0.1\,s}{s} \cdot e^{-0.1s}$$

and hence

$$\arg L_2(j\omega_c) = \arg(1+0.1\,j \cdot 7.5) - 90^\circ - \frac{18}{\pi} \cdot 7.5 \simeq -96.1^\circ$$

and hence $\varphi_m = 180^\circ - 96.1^\circ \simeq 83.9^\circ$ which definitely satisfies Requirement ii) thus concluding the design.

[28 Marks]

b) i) The open-loop transfer function is now

$$\widetilde{L}(s) = 12 \cdot \frac{1+0.1\,s}{1+24\,s} e^{-0.1s}$$

Answers

First, let us compute the crossover frequency ω_c:

$$|\widetilde{L}(j\omega_c)| = 1 \Longrightarrow 12 \cdot \frac{|1+0.1\,j\omega_c|}{|1+24\,j\omega_c|} = 1$$

$$\Longrightarrow \frac{\sqrt{1+\frac{1}{100}\omega_c^2}}{\sqrt{1+576\omega_c^2}} = \frac{1}{12} \Longrightarrow \omega_c = \frac{\sqrt{1-\frac{1}{144}}}{\sqrt{\frac{576}{144}-\frac{1}{100}}} \simeq 0.5\,\text{rad/s}$$

Now, we compute the phase margin:

$$\arg\widetilde{L}(j\omega_c) = \arg(1+0.1j\cdot 0.5) - \arg(1+24j\cdot 0.5) - \frac{18}{\pi}\cdot 0.5 \simeq -85.2^\circ$$

and hence $\varphi_m = 180^\circ - 85.2^\circ \simeq 94.7^\circ$ which is positive. Then, the closed-loop system under the action of the new controller $\widetilde{C}(s) = 1$ is asymptotically stable.

[7 Marks]

ii) Neglecting the time-delay block, the open-loop transfer function reads

$$\widetilde{L}'(s) = 12 \cdot \frac{1+0.1\,s}{1+24\,s}$$

The corresponding true Bode diagrams with their straight-line approximation are plotted in Figure 3.3 below.

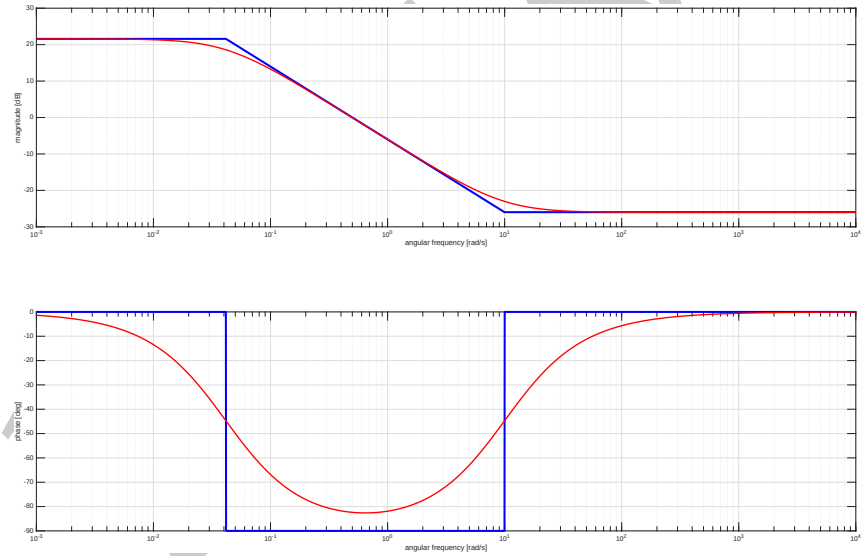


Figure 3.3 Bode diagrams of $\widetilde{L}'(j\omega)$.

Since $\widetilde{L}'(s)$ has no unstable poles, the gain is positive and the behaviour of the Bode diagram of the magnitude of the frequency response is monotonically decreasing crossing the 0dB axis once from the top to the bottom, the Bode criterion is applicable.

[5 Marks]

Answers